Reduced Dimensional Structures

In bulk metals and semiconductors electrons (or holes) are generally free to move in all three spatial directions. If this freedom is restricted in certain directions the dimensionality of the system becomes reduced. For instance, in a 2-dimensional system the electrons can only move in one plane and may not travel perpendicular to this plane. An example of a naturally occuring material showing quasi-2D behaviour is graphite where the resistance measured along the sheets is much lower than between sheets. Polymer sheets are another example of a 2D system as are electrons on the surface of liquid helium, but here we will concentrate on the 2D semiconductor systems found in MOSFETs, heterojunctions and *quantum wells*. By confining electron motion to a plane it becomes possible to control the actual position of the charges with surface gates and this forms the basis of the whole silicon electronics industry.

As well as confining the position of a particle, reducing the dimensions also changes the allowed energy levels, since the available momentum states are restricted. For a free particle of mass m the energy is $E = \frac{\hbar^2 \mathbf{k}^2}{2m} = \frac{\hbar^2 k_x^2}{2m} + \frac{\hbar^2 k_y^2}{2m} + \frac{\hbar^2 k_z^2}{2m}$, where $\mathbf{k} = (k_x, k_y, k_z)$ is the wavevector. In a 2D system with confinement in the z-direction, the term containing k_z will only have a few fixed values, corresponding to different *subband* energies E_i, while motion in the x-y plane is unrestricted. The energy dependence of the density of states also changes. In D dimensions $g(E) \sim E^{(D-2)/2}$, thus 2D is special with a constant density of states in each energy interval.

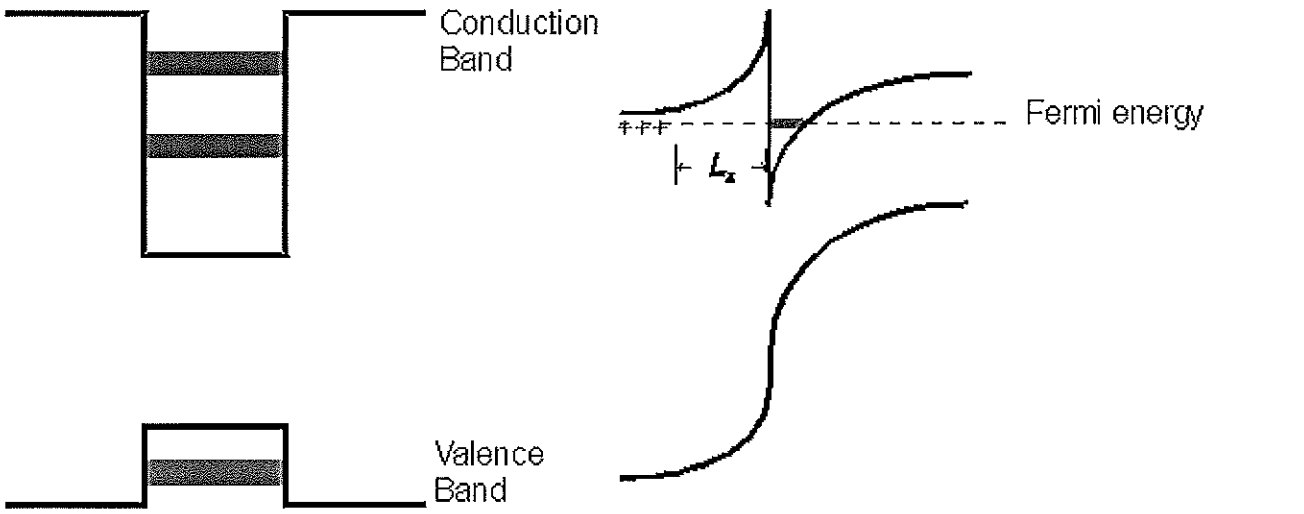
One dimensional systems, where motion is only along a line, also occur naturally e.g. transport along a polymer chain, and have been fabricated in semiconductors by advanced lithography techniques where they are known as *quantum wires*. Finally all possibility of movement may be removed to form a 0D system. The fabricated ones are called *quantum dots*, the natural version is an atom. But take care as one person's 0D system is another person's 3D system, it all depends on the length scales of interest! However a 0D system may be defined as one in which all the components of $\mathbf{k}$ are fixed and there are only discrete energy levels.

2-Dimensional Semiconductors

A 2D semiconductor system can be produced by changing the potential at the surface of a bulk semiconductor, with a metal gate, to form either an accumulation or inversion layer. In this way the carriers are trapped at the surface in an approximately

triangular potential well and form a 2-dimensional electron or hole gas (2DEG or 2DHG) depending on the doping of the material and the sign of the gate voltage. Silicon MOSFETs are made in this way (the oxide layer acts as an insulator between the metal gate and semiconductor) and were the first systems in which 2DEG physics was studied, including observation of the quantum Hall effect.

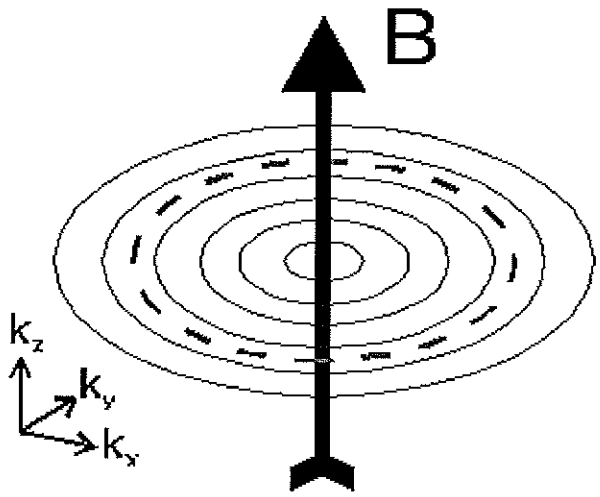
As the semiconductor-oxide interface in a MOSFET is rough a much higher mobility 2DEG can be formed by burying the interface within a crystal. Advanced growth techniques such as molecular beam epitaxy and chemical vapour deposition enable semiconductors to be grown one atomic layer at a time and abrupt interfaces to be formed between materials of different band gap e.g. GaAs and $Ga_{1-x}Al_xAs$ or Si and $Si_{1-x}Ge_x$. The figures below show the energy levels (not to scale) and how a 2DEG (in red) may be formed in the narrower gap material for a quantum well and a single heterojunction. The heterojunction is *modulation doped* which means the 2DEG is spatially separated from the ionised donors by an undoped spacer layer of thickness L_z. This greatly reduces impurity scattering and so increases the electron mobility.

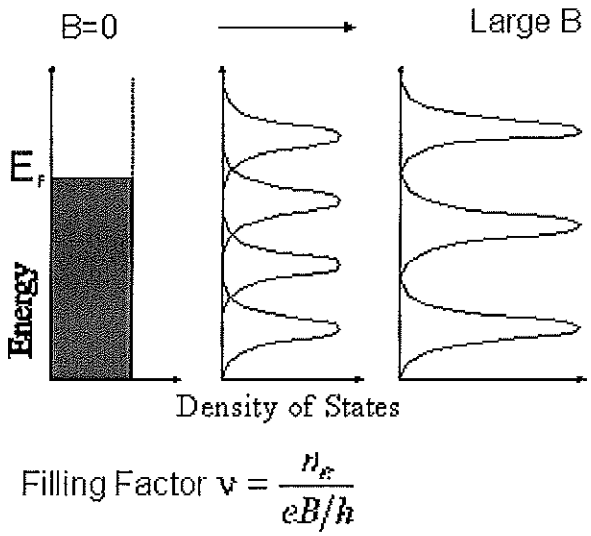


Magnetic Quantisation and Landau Levels

If a magnetic field is applied perpendicular to the plane of the 2DEG, the electron trajectories will be a set of circles around the lines of field. It can easily be shown that electrons perform these orbits at the cyclotron frequency $\boldsymbol{w_c}$**=eB/m***. The contribution form in-plane motion to the energy then becomes quantised in units of the cyclotron energy $E = E_i + \left(N + \frac{1}{2}\right)\hbar\omega_c$ and Landau levels (LL) are formed. The energy is thus completely defined by the subband index *i* and the Landau level

index *N*. As the magnetic field is increased the LL separation increases. In real materials scattering will broaden the LLs as illustrated below.





Filling factor

The number of states per unit area in each Landau level is equal to those originally within the range of one cyclotron energy in the 2D DOS, i.e. **eB/h** per LL. This is also the number of flux quanta threading unit area, which is no coincidence since the number of different electron wavefunctions that can be created in each LL is equal to the number flux quanta.

When particles are put into these states, the *filling factor* \nu is equal to the number of Landau levels filled. Clearly for partially filled LLs \nu will be a fraction. As the magnetic field is increased the filling factor will decrease since more particles can be put in each LL. The quantum limit is reached when only one LL is occupied.

The filling factor can also be usefully written as

$$\nu = \frac{\text{areal density of particles}}{\text{areal density of flux quanta}}$$



Last updated 18/02/97 by David R Leadley.

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